# COGNITIVE·TERRAIN

## PRACTITIONER'S MANUAL · RUN IT · BUILD IT · KNOW WHY




### RUN IT WITH A GROUP

Session plans from a lunch break to a multi-organisation exercise, and how to lead the debrief.

### BUILD YOUR OWN CAMPAIGN

The campaign and technology editors, balance checks and sharing.

### WHY IT WORKS THIS WAY

Design notes: what each mechanic models, and how it was arrived at.

EDUCATIONAL WARGAME · TURN-BASED
PLAYERS NEED ONLY THE FIELD MANUAL

CT · PM · 2026

FOR WHOEVER RUNS THE ROOM

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### HOW TO READ THIS MANUAL

| | |
|---|---|
| `4.5` | A section and rule of this manual. |
| `FM 10.3` | A rule in the Field Manual. This manual does not repeat the player rules; it points to them. |
| `DN 10.3` | The Designer's Note on Field Manual rule 10.3. |
| `EXAMPLE` | A rule worked through with real numbers. |

### FIRST TIME RUNNING IT?

Play **Typhoon Ines** on **Standard** once yourself, end to end, before you run it for anyone else. It takes about forty minutes. Then read Appendix A.

SECTION 1 · BEFORE YOU BRIEF

# 1.0 INTRODUCTION

## 1.1 WHO THIS IS FOR

Instructors, exercise designers and facilitators who run Cognitive Terrain for other people, build campaigns for their own audience, or need to explain why the game models cognitive warfare the way it does.

Players need only the **Field Manual**, which is also the guide inside the game. This manual assumes you have read it or played a campaign.

## 1.2 WHAT THE GAME TRIES TO GET ACROSS

- The target of cognitive warfare is the quality of the other side's decisions, not territory or attention.
- Reach is not effect: a viral post on the wrong area changes nothing that matters.
- Attack and defence happen at once, and open and closed societies are vulnerable in different ways.

## 1.3 USING THE TWO MANUALS

The Field Manual states every player rule once, by number. This manual covers what a facilitator needs on top: the campaigns, sessions, the White Cell, the debrief, the model's arithmetic and the editor. Section 9.0 gives the reasoning behind each rule, keyed to its Field Manual number.

## 1.4 WHAT THE GAME IS, AND IS NOT

> Cognitive Terrain is an **independent hobby project**, made for learning and discussion. It is free.
>
> It is **not doctrine**, and not an official, approved or endorsed product of any organisation. The authors of the research it draws on have not reviewed it.
>
> It has **not been independently assured or validated**. Its numbers are game rules, not measurements or predictions of real-world effects. Section 10.0 says which rules come from research, which are the designer's assumptions, and what has and has not been checked.
>
> Every campaign, town, outlet, person and event is **fiction**.

## 1.5 USING IT WITH A GROUP

Treat the game as a prompt for discussion, not as evidence. The most useful moments are usually the gaps: a side that wins on points but loses the debrief's verdicts, or a viral move that changed nothing [5.0].

SECTION 2 · FOUR FORMATS · SUGGESTED TIMINGS

# 2.0 RUNNING A SESSION

*The game runs offline on any computer with a browser, so the format is set by your time and your room, not by the technology. Timings are suggestions; adjust them to your group.*

## 2.1 FORMATS

### SOLO

ONE PLAYER · 30–45 MIN

1. Send the link or the HTML file with the Field Manual.
2. Ask for **Typhoon Ines** on **Recruit** with the tutorial first.
3. Ask them to bring a screenshot of the debrief and one turning point from the replay.

### CLASSROOM

GROUPS OF 2–3 · 50–90 MIN

1. 10 min: the decision loop (Field Manual 2.1) on the projector.
2. 30–60 min: each group plays one campaign on one screen, one driver, others advising.
3. 15–20 min: debrief as a room (5.0).

### PAIRED SEMINAR

BLUE vs RED · 60–90 MIN

1. Pair students; one plays Blue, one Red, on two computers in the same room.
2. They swap turn codes by chat (8.0). Sixteen weeks takes about an hour.
3. Debrief each pair, then compare how Red was played across pairs.

### DISTRIBUTED EXERCISE

SEVERAL ORGANISATIONS · DAYS

1. Build or choose a campaign and send the file to every cell first (7.1). Add a White Cell to steer it (4.0).
2. Cells play one week per day or per session, codes by email.
3. Collect debriefs and replays for a joint hot-wash.

## 2.2 DECIDING BEFOREHAND

**WHAT TO DECIDE BEFOREHAND**

**Plain or military terms.** Plain words suit mixed audiences; military terms (OODA, DQ, Seed Channels) suit staff courses. Players switch in the top bar.

**Campaign and difficulty.** See 3.0.

**Week in review on or off.** On for learning; off for a faster, harder exercise.

**KEEP IT MOVING**

Set Red's turn speed to **FAST** once players know what Red's moves look like, and result pop-ups to **KEY ONLY**. Both are in Settings and change nothing in the rules.

EXAMPLE

**A 90-minute classroom session.** Groups of two or three, one screen each, Typhoon Ines on Recruit.




Keep the last twenty minutes for the debrief (5.0), even if the campaign is unfinished.

# 2.3 SETTING UP THE ROOM

## 2.3 GETTING IT ONTO MACHINES

| OPTION | BEST WHEN |
|---|---|
| **Play in the browser** (playtest page) | Internet is available and nothing can be installed. |
| **One HTML file** | Locked-down or offline machines: copy it over and double-click. |

## 2.4 ON A PROJECTOR

Settings → **Chart Room** (the light scheme) reads best on washed-out projectors. Set **Interface size** to LARGER or LARGEST for the back row, and turn on **Colour-safe mode** so the sides differ by shape as well as colour.

## 2.5 PRIVACY

The game has no accounts, no analytics and no server; its only network request is for web fonts, and it works without them. Settings, saved two-player games, custom campaigns and a loaded voice pack stay in the browser on that machine. Clearing the browser's site data removes them.

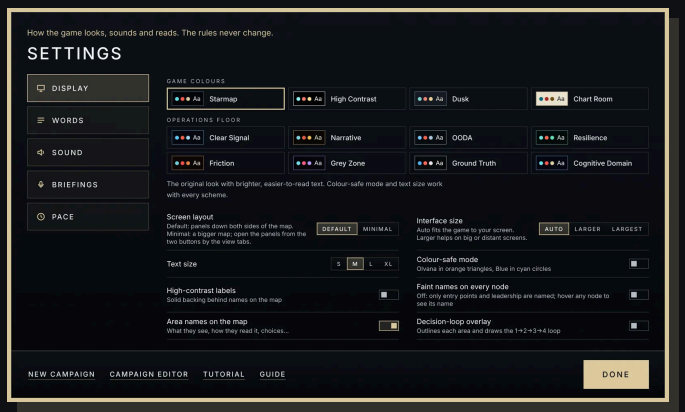



**FIG. 2.4** Settings. Chart Room, the light scheme, reads best on a washed-out projector; interface and text size are set separately.

## 2.6 IMMERSIVE BRIEFINGS IN A ROOM

With the optional voice pack, ▶ LISTEN in the week in review reads the week aloud as the broadcaster of the side ahead (Field Manual 15.3). It never starts by itself. In a room it works best on the presenter's machine with speakers.

1. Download **CognitiveTerrain-Voice-Emma.zip** (about 111 MB) once, before the session.
2. Settings → Immersive Briefings → **LOAD VOICE PACK...**, in the browser the session will use.
3. Press **HEAR A SAMPLE** to check the speakers.

It runs offline, on that machine only. The voice is synthetic speech from an open-source model; tell your audience so. Slower machines take a few seconds before the first line.

> **2.7 · ACCESSIBILITY CHECKLIST**
> Text size and interface size are separate settings. All text in every colour scheme passes 4.5:1 contrast. Reduced motion stops pulsing and drifting. Number keys pick moves; Esc cancels. Captions show whenever the narrator speaks.

SECTION 3 · EIGHT CAMPAIGNS · THREE DIFFICULTIES

# 3.0 CAMPAIGNS

*Every campaign uses the same rules. What changes is the setting, the opponent, the twist at the start and which areas the four crises test. Pick by what you want the room to argue about.*

## 3.1 CHOOSING ONE

| CAMPAIGN | VS | TWIST AT THE START | CRISES TEST | GOOD FOR |
|---|---|---|---|---|
| **Typhoon Ines** | Olvana | None | Notice · Interpret · Act · Trust | First games; the core loop |
| **The Valmora Vote** | Olvana | Both TRUST areas weakened | Interpret · Notice · Decide · Trust | Elections, polarisation |
| **Collision at Varrin Reef** | Olvana | Both officials' entry points weakened | Notice · Act · Act · Decide | Crisis diplomacy, competing videos |
| **Fever in Port Sabre** | Olvana | Rumours spread faster for both; both NOTICE areas strained | Interpret · Trust · Act · Decide | Infodemics, public health |
| **Winter Grid** | Donovia | Blue's FOLLOW-THROUGH strained; 4 dormant narratives | Notice · Decide · Act · Trust | Energy coercion |
| **The Lake Referendum** | Donovia | Donovia already in Blue's INTERPRET; TRUST weakened | Interpret · Notice · Decide · Trust | Referendums, planted premises |
| **The Silent Survey Ship** | Donovia | Both DECIDE areas strained | Notice · Act · Interpret · Decide | Ambiguity, alliance choices |
| **The Vell Valley** | Donovia | 5 dormant narratives; Blue's NOTICE and TRUST strained | Interpret · Act · Trust · Decide | Annexation and its cover story |

## 3.2 DIFFICULTY

| LEVEL | WHAT CHANGES |
|---|---|
| **Recruit** | Red is slower, makes more mistakes and earns 2 fewer points a week. For first games and short sessions. |
| **Standard** | A steady, sensible Red. Campaigns are balanced at this level. |
| **Veteran** | Red plans two moves ahead. No extra budget, no hidden information. For experienced players and competitions. |

### 3.3 · MATCH THE OPPONENT TO YOUR TOPIC

**Olvana** is designed around patience against nerve: a closed state that works one area at a time from inside.

**Donovia** is designed to show that the aim can be doubt and inaction rather than persuasion: floods, planted premises, shows of force.

Field Manual 13.3 and 13.4 set out each one's rules and sources.

### SAME MAP, FAIR COMPARISON

Groups playing the same campaign get different maps unless they start from the same seed. For a like-for-like comparison, run it as a two-player game from one Blue code (8.0).

SECTION 3 · FOUR CAMPAIGNS · ALL ORIGINAL FICTION

# 3.4 THE OLVANA CAMPAIGNS

*Same rules, different pressure. Each campaign has its own crises, its own tested areas and its own starting conditions (FM 13.3 sets out how Olvana fights).*

## TYPHOON INES

DISASTER RELIEF · THE KALVARA ISLANDS

A typhoon flattens a small island republic that trades with both sides. Over sixteen weeks of relief and rebuilding, Kalvara chooses whom to trust.

TWIST None. The best first campaign.

| | | |
|---|---|---|
| W04 | **The Relief Flotilla** | NOTICE |
| W08 | **The Blackout** | INTERPRET |
| W12 | **The False Alarm** | ACT |
| W16 | **The Reconstruction Compact** | TRUST ×2 |

## THE VALMORA VOTE

A SNAP ELECTION · VALMORA

A partner democracy holds a snap election. Olvana quietly favours the candidate who would cancel the Blue trade pact.

TWIST Both TRUST areas start weakened (Blue's more), and AI Synthetic Media arrives first.

| | | |
|---|---|---|
| W04 | **The Leaked Audio** | INTERPRET |
| W08 | **The Ballot Rumor** | NOTICE |
| W12 | **Count Night** | DECIDE |
| W16 | **Transfer of Power** | TRUST ×2 |

## COLLISION AT VARRIN REEF

MARITIME INCIDENT · TELUGA

An Olvanan cutter and a fishing boat from Teluga, a small Blue ally, collide at a disputed reef. Two fishermen are missing.

TWIST Both sides' Government Contacts start weakened, so the fight over envoys starts on day one.

| | | |
|---|---|---|
| W04 | **Two Videos** | NOTICE |
| W08 | **The Missing Crew** | ACT |
| W12 | **The Drills** | ACT |
| W16 | **The Tribunal Ruling** | DECIDE ×2 |

## FEVER IN PORT SABRE

AN OUTBREAK · PORT SABRE

A fast-spreading virus hits a trading city-state that depends on both sides. Rumours move faster than the virus.

TWIST Infodemic. Held nodes push harder on their neighbours and both NOTICE areas start strained.

| | | |
|---|---|---|
| W04 | **Patient Zero** | INTERPRET |
| W08 | **The Bad Batch** | TRUST |
| W12 | **The Forged Order** | ACT |
| W16 | **The Health Accord** | DECIDE ×2 |




**REPLAY VALUE**

Every campaign draws a fresh brain map, and the headlines come from a large story bank written for each scenario, so you can play many games before stories repeat.

**TRY THIS**

Play the same scenario on **Standard** twice: once attacking every turn, once defending first. Compare the two debriefs.

SECTION 3 · FOUR CAMPAIGNS · ALL ORIGINAL FICTION

# 3.5 THE DONOVIA CAMPAIGNS

*Donovia fights by its own rules (FM 13.4). Its four campaigns are set in fictional countries from the U.S. Army's training environment; every town, outlet, person and event in them is invented.*

## WINTER GRID

A COLD SNAP · BOTHNIA

A record cold snap strains the grid of a small northern democracy that buys a third of its winter power from Donovia. Donovia offers to keep the lights on, for a price.

TWIST Blue's ACT area starts strained; 4 dormant narratives.

| | | |
|---|---|---|
| W04 | **The First Blackout** | NOTICE |
| W08 | **The Price of Heat** | DECIDE |
| W12 | **The Substation Fire** | ACT |
| W16 | **The Winter Compact** | TRUST ×2 |

## THE LAKE REFERENDUM

A REFERENDUM · FRAMLAND

Framland votes on rebuilding a dam on the lake it shares with Donovia. Donovia wants the joint project; many Framlanders do not.

TWIST Donovia starts with a foothold in Blue's INTERPRET; Blue's TRUST starts weakened.

| | | |
|---|---|---|
| W04 | **The Engineers' Report** | INTERPRET |
| W08 | **Ten Thousand Neighbours** | NOTICE |
| W12 | **Debate Night** | DECIDE |
| W16 | **Referendum Day** | TRUST ×2 |

## THE SILENT SURVEY SHIP

A SHIP GOES QUIET · ARNLAND

An Arnland research ship stops transmitting near Donovian waters, while Arnland decides whether to join Blue's alliance.

TWIST Both DECIDE areas start strained.

| | | |
|---|---|---|
| W04 | **Radio Silence** | NOTICE |
| W08 | **Snap Drills** | ACT |
| W12 | **Three Stories** | INTERPRET |
| W16 | **The Inquiry** | DECIDE ×2 |

## THE VELL VALLEY

ANNEXATION · GORGAS

Donovia has said for years that Donovian speakers in a Gorgas valley need its protection. Now armed men without insignia are on the valley roads.

TWIST 5 dormant narratives; Blue's NOTICE and TRUST start strained.

| | | |
|---|---|---|
| W04 | **The Protection Claim** | INTERPRET |
| W08 | **Men Without Insignia** | ACT |
| W12 | **The People's Vote** | TRUST |
| W16 | **The New Map** | DECIDE ×2 |

### FICTION, WITH REAL METHODS

No campaign retells a real event. What comes from research is **how Donovia operates**: flooding the zone, planting premises, dormant narratives, shows of force, denial and staged legitimacy. Each campaign names its sources on the start screen (**How Donovia fights**). Donovia, Bothnia, Framland, Arnland and Gorgas are countries in the U.S. Army TRADOC Decisive Action Training Environment (DATE); they are fictional training countries.

SECTION 4 · STRUCTURED SESSIONS · A THIRD PLAYER WHO STEERS

# 4.0 THE WHITE CELL

*In a structured session a facilitator joins as the **White Cell**. It sees both minds, and once a week, after Red and before Blue, it can adjust the game: run a story, change a budget, move a crisis, strike a move. It is the game's version of the control cell in a staff exercise.*

## 4.1 ORDER OF PLAY

**BLUE** Plays the week — Code to Red → **RED** Plays the week — Code to White Cell → **WHITE CELL** Watches both, then steers — Code to Blue → **BLUE** Reads the changes, plays — Next week begins

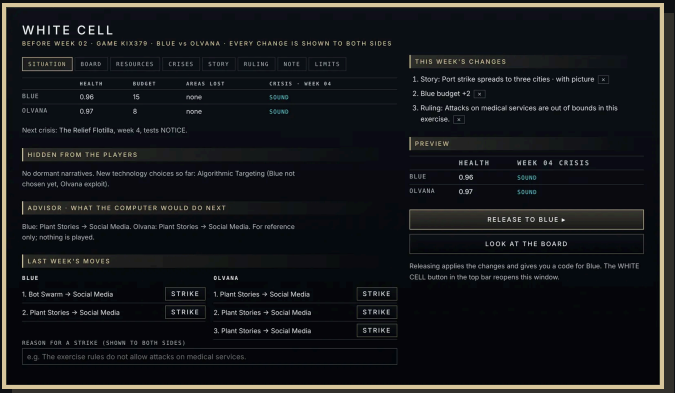



**FIG. 3** The White Cell console before week 2: both sides' health, budget and next crisis; what is hidden from the players; last week's moves with STRIKE buttons; this week's changes and their effect on the right.

**ONLY WHEN YOU CHOOSE IT**

Ordinary two-player games are unchanged. The White Cell exists only when Blue starts a **NEW STRUCTURED SESSION · WITH WHITE CELL**.

## 4.2 SETTING ONE UP

1. **Blue starts it.** TWO PLAYERS → NEW STRUCTURED SESSION · WITH WHITE CELL. Blue plays week 1 and sends the code to Red.
2. **Red joins** with Blue's code, as in any two-player game, and sends its code to the White Cell, not to Blue.
3. **The White Cell joins** with Red's first code: TWO PLAYERS → JOIN AS WHITE CELL. Both week 1 turns replay and the console opens.
4. **Release.** Queue changes, or none, and press RELEASE TO BLUE. Send the code to Blue. Week 2 begins.

## 4.3 WHAT EACH PLAYER SEES

**WHAT THE WHITE CELL SEES**

Both minds with no fog, both budgets, every move as it replays, Red's secret technology choices and Blue's dormant narratives. It plays no moves of its own.

The **WHITE CELL** button (where END TURN usually is) reopens the console; **LOOK AT THE BOARD** closes it to study the map.

**WHAT THE PLAYERS SEE**

Nothing is secret. Every story, ruling and change appears in a White Cell notice (4.7): Blue sees it before its turn, Red when Blue's code arrives. It is also in the log, the debrief and the replay.

# 4.4 THE CONSOLE

*Eight tabs, one queue, almost all of it buttons. Each change is listed on the right as you add it, with a preview of both sides' health and the next crisis. Remove any item before you release; nothing happens until you do.*

**SITUATION**
Health, budgets, the next crisis, hidden facts, and what the computer would play next. STRIKE a move here.

**BOARD**
Pick a node on the map, then one click to shift control, set or clear a status, or plant a dormant narrative.

**RESOURCES**
One click to add or take budget points, or give either side a move more or fewer this week.

**CRISES**
Move, retarget or double a coming crisis, add a new one, or schedule a new technology.

**STORY**
A breaking news item, with an optional picture. Blue reads it before its turn, Red with Blue's next code.

**RULING**
An adjudication the players must follow, with your reason.

**NOTE**
A teaching note kept out of play; it appears in the debrief and replay.

**LIMITS**
Your own caps: largest shift and budget change; switch moves, crises or strikes off.

## 4.5 CHANGES AND THEIR LIMITS

| CHANGE | LIMIT THE GAME KEEPS | YOUR DEFAULT LIMIT |
|---|---|---|
| Shift a node's control | Stays within −100 to +100 | Up to 25 per change |
| Watched, prebunked, framed | 0 to 4 weeks; 0 clears it | — |
| Dormant narrative | Blue's mind only | — |
| Budget | Stays within 0 to 18 | Up to 3 per change |
| Moves this week | One more or one fewer; Blue's at once, Red's when its turn starts | Can be switched off |
| Crisis | Not in the past, not in a week that already has one | Can be switched off |
| New technology | A future week without one | Can be switched off |

Set your own limits in the **LIMITS** tab; they change only what the console offers.

## 4.6 STRIKING A MOVE

A strike removes one of the past week's moves. Every copy rebuilds the week without it, and both sides are told why. Strikes can be switched off in LIMITS.

## 4.7 WHO SEES WHAT, AND WHEN

Every change, story and ruling is shown to both sides. Blue sees the White Cell notice before its turn; Red sees it when Blue's code arrives. Teaching notes are the exception: they appear only in the debrief and the replay.

**WORKED EXAMPLE**

The next page follows one real week at the console, from finding the pressure on the map to the notice Blue reads.

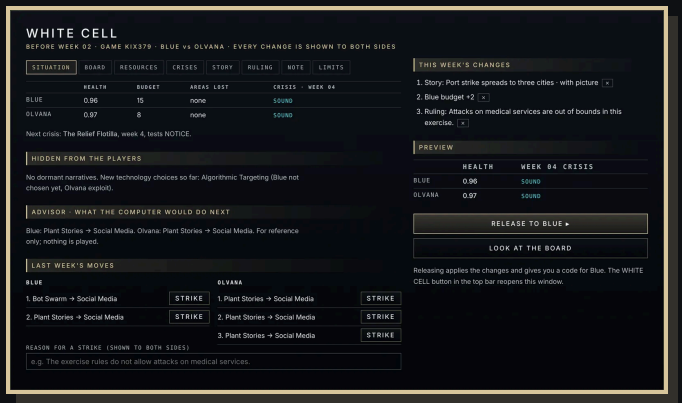



**FIG. 3** The SITUATION tab: health, budgets, the next crisis, what is hidden from the players, and last week's moves with STRIKE buttons.

SECTION 4 · EXAMPLE · ONE WEEK AT THE CONSOLE

# A WEEK AT THE CONSOLE

*Typhoon Ines, before week 2. Olvana spent its first turn hammering Blue's Social Media. The facilitator wants the room to feel an outside shock and to see how Blue responds to pressure on resources. Every picture is the real game.*

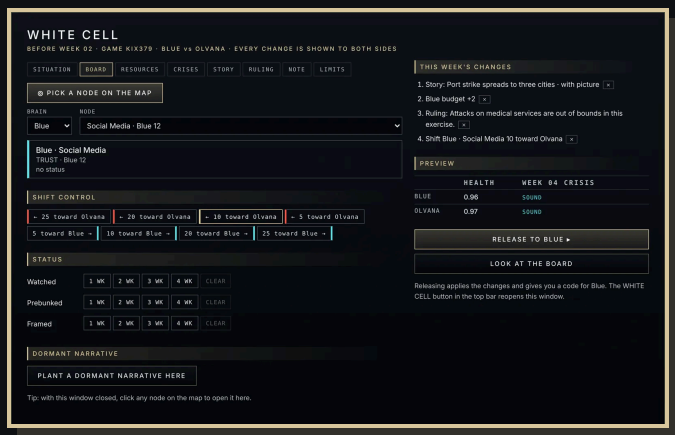



**STEP 1 · FIND THE PRESSURE** The White Cell sees Blue's mind with no fog. Social Media is down to Blue 12, contested, after Olvana's Plant Stories. **PICK A NODE ON THE MAP** closes the console; the bar at the bottom says what to do. Click Social Media.

## THE PLAN

1. **Story:** "Port strike spreads to three cities", with a picture.
2. **Board:** shift Blue's Social Media 10 toward Olvana, as the clip spreads online.
3. **Resources:** Blue +2 budget, to see where it spends.
4. **Ruling:** attacks on medical services are out of bounds.

> FACILITATOR'S NOTE Pair every board change with a story, so players know why it happened.




**STEP 2 · QUEUE THE CHANGES** The console reopens on Social Media. One click on **10 toward Olvana** queues the shift. The list on the right holds all four changes; the preview shows both sides still SOUND for the week 4 crisis. Remove any item with ×. Then **RELEASE TO BLUE** .




**STEP 3 · WHAT THE PLAYERS SEE** Blue loads the code and reads this before its turn: the story and picture, the ruling, and every change in plain words. Red reads the same notice when Blue's week 2 code arrives (4.7).

# 4.8 STEERING THE EXERCISE

## 4.8 STORIES AND PICTURES

A story is a short breaking-news item with an optional picture. Drop, paste or browse for an image: the game shrinks it to fit inside the turn code, so it travels by email or chat like any other code. A picture that cannot be made small enough is refused with a message.

## 4.9 RULINGS AND NOTES

A **ruling** is an adjudication the players must follow, with your reason. A **note** is a teaching note kept out of play for the debrief. Neither changes the board.

## 4.10 ENDING THE SESSION

When Red's last turn ends the game, Red's final code comes to you; release it and send the last code to Blue so every copy ends together. The debrief carries the White Cell's record and notes.

**POINT AND CLICK**

Click any node on the map and the console opens on it, ready for a shift or a status. **LOOK AT THE BOARD** closes the console; the **WHITE CELL** button brings it back.

## 4.11 SUGGESTED INJECTS

| IDEA | TRY |
|---|---|
| Events outrun preparation | Bring the next crisis forward a week, or make it test the area a side has neglected. |
| An outside shock | A story plus a small control shift on the matching node, e.g. a viral clip and a nudge on Social Media. |
| Resource pressure | Cut a side's budget by 2 before a crisis and watch what it gives up. |
| Rules of engagement | A ruling, then strike any move that breaks it. |
| A losing side recovers | One extra move for a week; ask in the debrief what it did with it. |

> FACILITATOR'S NOTE Say in the debrief why you changed what you changed.

# 5.0 THE DEBRIEF

*The game ends with a neutral debrief and a replay of both minds week by week (FM 12.4). Most of the learning happens here, so leave time for it.*

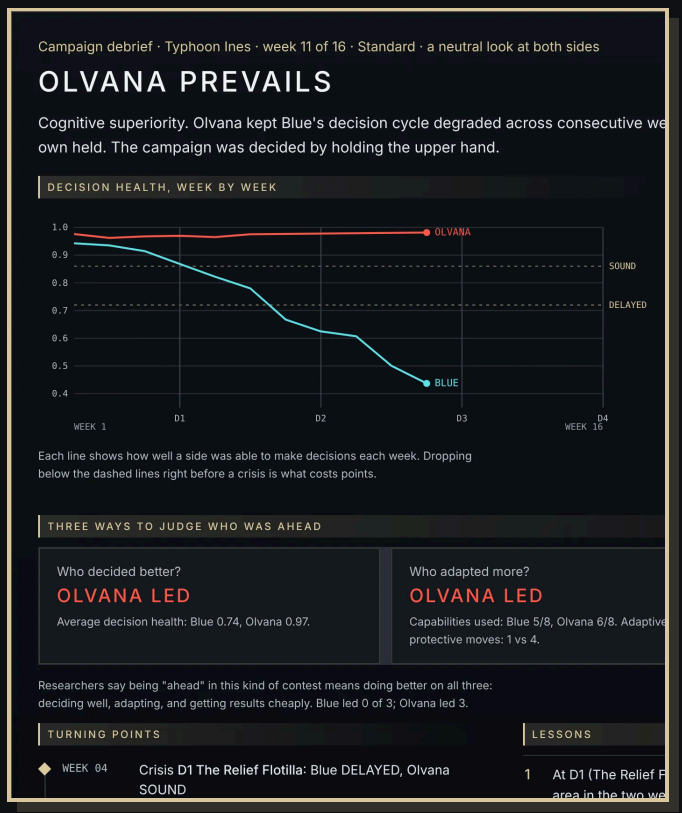



**FIG. 1** The debrief: crisis points and health week by week, followed by three verdicts, turning points and lessons further down.

## 5.1 THE THREE VERDICTS

| VERDICT | MEASURES |
|---|---|
| **Who decided better** | Average Decision Health across the campaign, week by week. |
| **Who adapted more** | How many of its capabilities (eight, nine for Donovia) each side used, and how many adaptive or protective moves. |
| **Who got more for less** | Influence moved per point spent attacking. |

A side can win on points and lose two of the three verdicts. That gap is often the best conversation in the room.

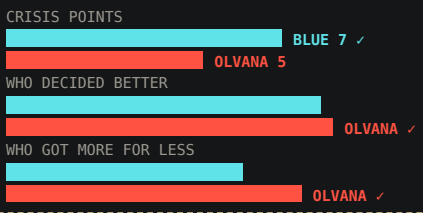
## 5.2 THE REPLAY

Step or play through every week: both minds, that week's moves, and the turning points (▼ ▲) and crises on the strip. Stop on a turning point and ask what each side knew then.

EXAMPLE

**An illustration of a split result.** Blue wins the campaign 7–5 on crisis points. But Olvana's average Decision Health across the sixteen weeks was higher, and it moved more influence for each point it spent. Two of the three verdicts go to the side that lost.

Ask the room: what did Blue do that won the crises, and what did it cost?




## 5.3 QUESTIONS

### QUESTIONS THAT WORK

- Which crisis did you lose, and when did you first know it was coming?
- Find a move with a great spread score that changed nothing. Why?
- When did defending your own mind matter more than attacking theirs?
- What did Red seem to be trying to do? What told you?
- Which of Blue's advantages as an open society did you use, and which did you lose?
- What would you tell a decision-maker in the real version of this crisis?

SECTION 6 · HOW A MIND IS SCORED

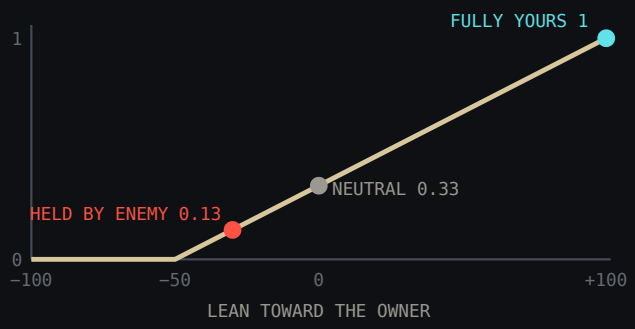
# 6.0 THE MODEL IN NUMBERS

*Every number the game uses to score a campaign, in plain words first and then as a formula, with a worked example. Nothing here is hidden from players: the side panel and the week in review show the same sums.*

## 6.1 AREA HEALTH

Each area's health is how much of it the owner still controls, from 0 to 1. Nodes that are slipping get part credit, so the score moves smoothly rather than in jumps. A node fully yours counts 1; a neutral node about a third; a node the enemy holds hard counts nothing.

```
area = average over its nodes of
clamp( (lean + 50) ÷ 150, 0, 1 )
```




## 6.3 WHAT MOVES THE NEEDLE

How much each change is worth, to Decision Health and to the next crisis:

| CHANGE | DECISION HEALTH | NEXT CRISIS |
|---|---|---|
| Tested area +0.10 | +0.01 to +0.04 | **+0.05 to +0.06** |
| Any other area +0.10 | +0.01 to +0.04 | +0.01 to +0.02 |
| TRUST +0.10 | about +0.02 | about +0.01 (more if it is tested) |
| Lose an area | −0.06 and more | −0.04; −0.12 and more if tested |

In short: the area the next crisis tests is worth three to five times any other. That is why Looking ahead names it.

## 6.2 DECISION HEALTH

Decision Health adds up the four loop areas, weighted by how much each side relies on them. The total is then scaled by trust, and 0.06 comes off for every area lost.

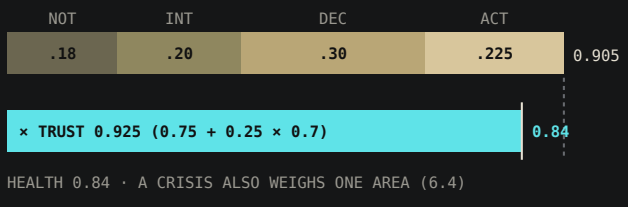
```
areas = w·NOTICE + w·INTERPRET + w·DECIDE +
w·ACT
DQ = areas × ( 0.75 + 0.25 · TRUST )
     − 0.06 × areas lost − flood
```

| WEIGHT | NOT | INT | DEC | ACT |
|---|---|---|---|---|
| BLUE | 0.20 | 0.25 | 0.30 | 0.25 |
| OLVANA · DONOVIA | 0.15 | 0.20 | 0.40 | 0.25 |

Trust can take a quarter off everything: with trust gone, the most a side can score is 0.69. **Flood** (Donovia only): −0.02 for each of its attacks that lands in a week beyond the first two (one more is free per Lessons Learned level), at most −0.06, until Donovia's next turn.

EXAMPLE

Blue's areas: NOTICE 0.9, INTERPRET 0.8, DECIDE 1.0, ACT 0.9; TRUST 0.7; nothing lost. Each area adds its share; trust then trims the total.




HEALTH 0.84 · A CRISIS ALSO WEIGHS ONE AREA (6.4)

# 6.4 CRISES AND WINNING

## 6.4 A CRISIS

A crisis mixes a side's general health (60%) with the health of the one area it tests (40%). Losing that area costs another 0.08. The result falls in one of three bands.

```
crisis = 0.6 · DQ + 0.4 · tested area
− 0.08 if the tested area is lost
```

| ERROR | DELAYED | SOUND |
|---|---|---|
| below 0.72 · 0 pts | 0.72+ · 1 pt | 0.86+ · 3 pts |

**Plus 1** to a side whenever the other side scores ERROR. A crisis that counts double (week 16 in the built-in campaigns) doubles everything, bonus included.

The "decides in ... hours" shown in the game is 6 ÷ (Decision Health, or the crisis score at a crisis)², rounded, and never more than 267. It is a way of showing health, not a separate rule.

## 6.5 WINNING

Most crisis points after week 16 wins. **Tied?** The side whose Decision Health is more than 0.03 higher wins; otherwise it is a draw. **Knockout**, earlier: the other side has lost at least 2 areas, and at least 2 more than you, at the end of 2 weeks running.

> FACILITATOR'S NOTE Ask players to predict a crisis result before it lands; then compare.

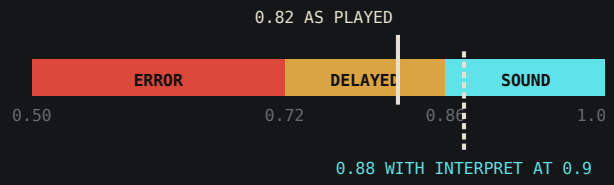
EXAMPLE

**The same Blue, one crisis, two outcomes.** Blue's Decision Health is 0.84 (6.2) and the crisis tests INTERPRET, which is at 0.8:

`0.6 × 0.84 + 0.4 × 0.80 = 0.82 → DELAYED, 1 point`

Had Blue spent the two weeks before lifting INTERPRET to 0.9, its Decision Health would be 0.86, and:

`0.6 × 0.86 + 0.4 × 0.90 = 0.88 → SOUND, 3 points`




One area, raised by a tenth, is worth 2 points here. That is the lesson of 6.3 in a single crisis.

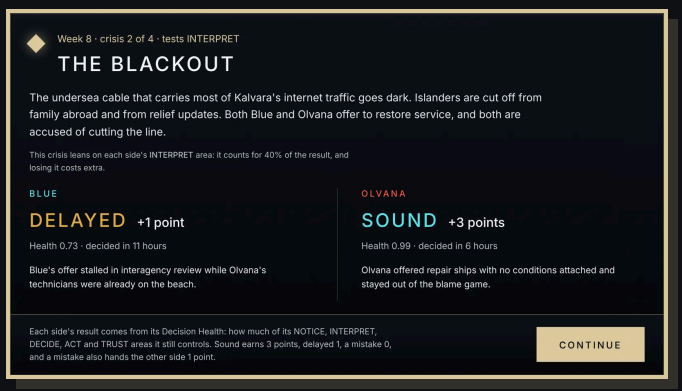



FIG. 6.4 How players see a crisis result: each side's band, the points, and what each side actually did.

# 6.6 MOVES AND ODDS

*How hard a move hits, how often it fails and what comes in each week. The Field Manual explains each in play (FM 7.4–7.7, 9.0); here is the arithmetic in one place.*

## 6.6 SPREAD SCORE AND GRADE

Every attack except Show of Force rolls an audience, scores it with the research's engagement formula, and turns the score into a grade that multiplies the push.

```
score = ( 0.1·views + 0.3·likes +
0.7·comments + 1·shares ) ÷ posts
```

| GRADE | SCORE | PUSH × |
|---|---|---|
| F · E · D | under 3 · 3 to under 4 · 4 to under 10 | 0.25 · 0.45 · 0.7 |
| C · B | 10+ · 100+ | 1.0 · 1.2 |
| A · A+ | 1,000+ · 10,000+ | 1.4 · 1.6 |

Reach is rolled on a ten-times scale, so from grade C upward ×10 is one grade. **×10:** their TRUST lost; each technology you used. **÷10:** a Watch Team on the target. **Smaller:** entry point ×2.5; their News Media held ×3.2; Red into Blue ×1.25; Blue into Olvana ÷1.4 (Donovia ÷1.25). Worked example: FM 7.7.

## 6.7 PUSH

The push is the card's power, made bigger or smaller by the area, the audience and the node's state.

```
push = card power × area fit × grade
× 0.5 if prebunked (0.45 for Donovia's
attacks)
× 1.4 if framed · × 2 if it wakes a dormant
narrative
× 1.1 for Blue's attacks on Donovia's TRUST
```

Card power: Plant Stories 20 (×1.5 on an entry point), Frame 16, Bot Swarm 12 plus 7 on every link (12 if you hold their Social Media, or used Social Bots or Autonomous AI Agents and the other side did not prepare), Repost 10, Show of Force 16 with no grade. Official Statement restores 14 (Blue ×1.6 while TRUST holds; ×1.5 on a watched node; ×0.5, with no Blue bonus, once TRUST is lost). Worked example: FM 7.4.

## 6.8 ODDS OF FAILURE

Chances add up. A failed attack pushes back toward the defender.

| MOVE FAILS IF… | CHANCE |
|---|---|
| Any attack | 8% |
| + target watched | +15% |
| + target prebunked | +8% |
| + Blue posting into Olvana / Donovia | +2% / +1% |
| Show of Force | +4% |
| Official Statement ignored | 6% (+20% if TRUST lost) |

A failed attack pushes the node back 6 toward the defender (9 when Donovia's fail); a failed Show of Force also pushes each neighbour back 4. With ACT lost, 1 move in 4 misfires before any of this is rolled.

## 6.9 EACH WEEK

**Budget:** 6 + 1 per enemy node held (Top Leadership 2) + 2 for their Government Contacts; at least 2 a week; at most 18 held. Recruit Red −2; Donovia +1. Extra move: Blue 2, Red 5, once a turn after the last move, only if 2 would be left.

**Recovery:** each node +1 toward its owner (+1 per Lessons Learned level, half for Red), halved with TRUST lost. Blue's TRUST nodes +2 more; Red's entry points ×1.2 (Donovia's ×1.1); Red's nodes leaning Blue ×0.35. None while framed or once held by the enemy. Any neighbour leaning 50+ the other way pushes 5 (6 in Fever in Port Sabre; half if prebunked), and +1 more toward the owner if any neighbour leans 50+ its way. Worked example: FM 9.5.

# 7.0 BUILDING CAMPAIGNS

*Set the scene for your own audience, place the crises and new technologies on the calendar, tune how strong each side starts, then test it, check its balance and share it. No code involved.*

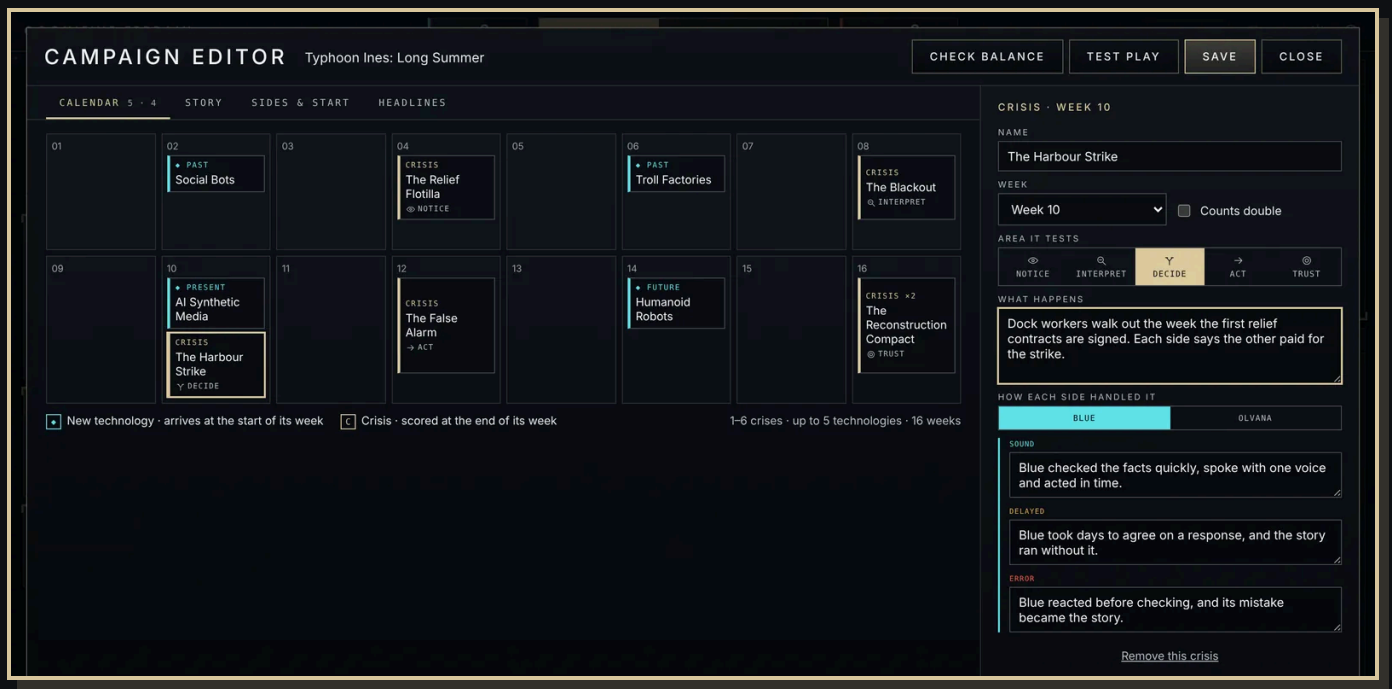



**FIG. 5** The campaign editor, editing a copy of Typhoon Ines with a new crisis in week 10. Click a crisis or technology on the calendar to edit it on the right; + adds one in that week.

## 7.1 MAKE ONE

1. On the start screen press **CAMPAIGNS** (or ⚙ → **CAMPAIGN EDITOR**).
2. **+ NEW CAMPAIGN**, or **COPY & EDIT** a built-in one. Copying is the fastest way to a balanced start.
3. Choose the opponent (Olvana or Donovia), write the setting and the headlines' places and outlets, and set any twist at the start.
4. Place 1 to 6 crises and up to 5 new technologies (7.2).
5. **CHECK BALANCE**, then **TEST PLAY**.

> **SHARING**
>
> Save a **.campaign.json** file or copy a code that starts `CTC1-`. Campaigns are plain text and cannot run code, so they are safe to pass around. Campaigns live in the browser that made them: keep the file.

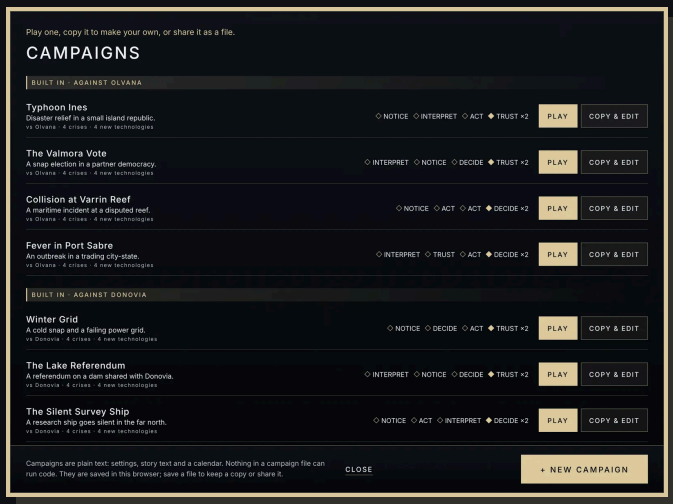



**FIG. 6** The campaign library: play a campaign, or copy one to edit.

# 7.2 CRISES AND TECHNOLOGIES

## 7.2 WRITING A CRISIS

A crisis is a moment that tests one area of both minds. For each one you write what happens, choose the area it tests, and write how each side handles it at each result: SOUND, DELAYED and ERROR. Players read your text at the moment of truth, so keep each line to one or two sentences about a decision, not an outcome.

> **Pick the area by the decision.** A flood of rumours tests NOTICE; a contested video tests INTERPRET; a deadline tests DECIDE; a delivery that must happen tests ACT; a scandal tests TRUST.

In the built-in campaigns the last crisis counts double; in your own, tick **Counts double** on any crisis. Spread the tested areas so no single defence wins the campaign.

### CHECKING BALANCE

**CHECK BALANCE** plays your campaign 40 times, computer against computer, and reports how often each side wins. 7.4 explains how to read the result.

**ORIGINAL FICTION**

The built-in campaigns invent every place, outlet, person and event. Doing the same keeps a session about methods rather than about a real event, and keeps it usable in any room.

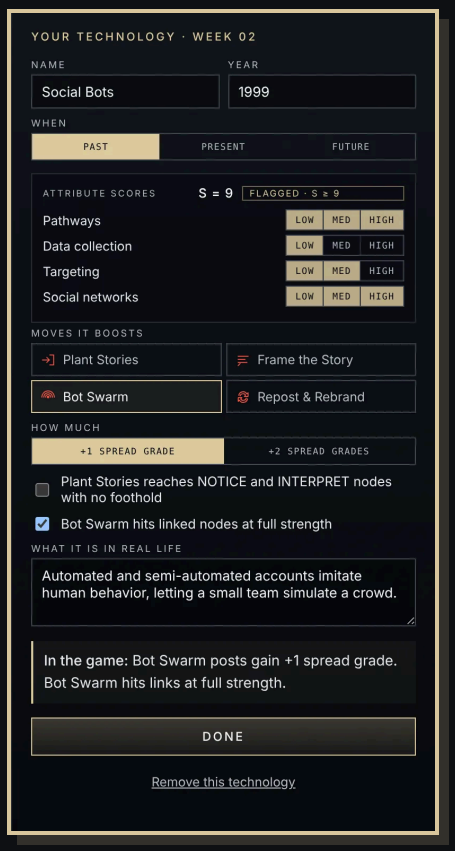



**FIG. 7** Building a technology: name, year, era, the four attribute scores and their total S, and which moves it boosts.

## 7.3 TECHNOLOGIES

Pick from 18 preset technologies, past, present and speculative, or **BUILD YOUR OWN**. Each is scored 1 to 3 on up to four attributes from Rushing, Umeokolo & Xu (2025): new pathways, data collection, targeting and social networks. A total of 9 or more is the paper's mark of a disruptive innovation. Where the paper gives no date or score, the editor marks the figure as illustrative or the designer's estimate.

# 7.4 READING A BALANCE CHECK

*CHECK BALANCE plays your campaign 40 times, the computer against itself, and reports how often Blue wins. Forty games is enough to spot a lopsided campaign, but not enough to tell 45% from 55%.*

## 7.4 WHAT THE NUMBER MEANS

The editor calls 25–65% Blue wins a **fair match**. The built-in campaigns sit between about 35% and 60%.

Any count of 40 games carries luck. A result is only good to about ±15 points either way: a campaign that shows 35% might really be close to 50%. The chart shows the likely true range for three results.

## 7.5 GETTING A FIRMER ANSWER

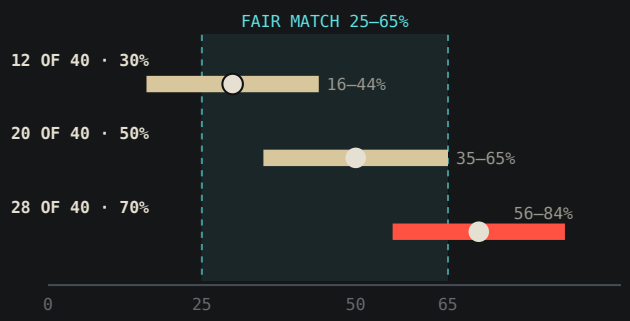
Run **CHECK BALANCE** again and add up the wins. Two runs (80 games) narrow the range to about ±11 points; four runs (160 games) to about ±8.

## 7.6 IF IT IS LOPSIDED

- Soften the twist at the start, or give it to both sides.
- Move a crisis to an area the stronger side has not prepared.
- Spread the tested areas, so no single defence wins every crisis.
- Change one thing at a time, then check again.

EXAMPLE

**Three results, three ranges.** Each bar is a 95% range (normal approximation); the shaded band is the editor's fair match.



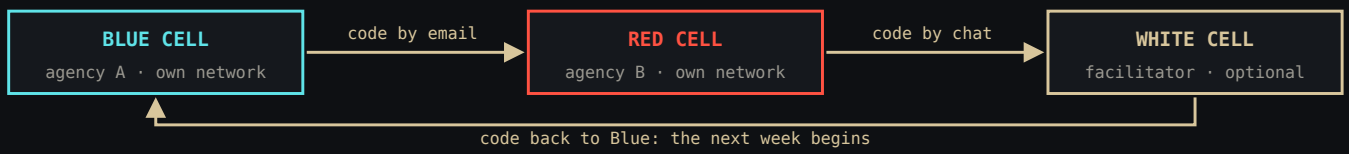

30% looks low, but its range reaches 44%: run it again before you change anything. 70% is outside the fair band even at its low end: that campaign favours Blue.

FACILITATOR'S NOTE Balance is tested computer against computer; people will play it differently.

SECTION 8 · TWO PLAYERS · NO SERVER · NO SHARED NETWORK

# 8.0 ACROSS ORGANISATIONS

*Two-player games need nothing but a way to send a line of text (FM 14.0). That makes the game usable between cells, units or agencies that cannot share a network or install software.*




One week of a structured session between organisations. Without a White Cell, Red's code goes straight back to Blue.

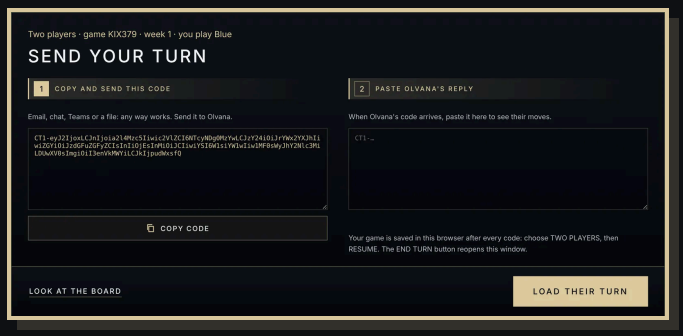



**FIG. 8** A turn code with a COPY button, and a box for the reply.

### 8.1 · WHAT A CODE CARRIES

The game and its map seed, the campaign, the week, the side and that side's moves. Each copy replays the other's moves and checks the board matches; if it ever does not, the log says so. Codes contain no personal data.

## 8.2 CHECKLIST

1. **Same version.** Codes work only between builds with the same minor version (0.18.x with 0.18.x), and the game cannot check this for you: a mismatch shows only as a failed board check. Send everyone the same download.
2. **Same campaign.** For a custom campaign, Red and any White Cell import the file before their first code (7.1). The code checks it.
3. **Assign sides.** Blue starts the game; Red joins with Blue's first code. The campaign decides whether Red is Olvana or Donovia. To steer it yourself, start a structured session and join as the White Cell (4.0).
4. **Set a rhythm.** One week per session or per day keeps a sixteen-week campaign moving.
5. **Collect the debriefs.** Each side sees the same debrief at the end; Red sends one final code so Blue's game ends too (in a structured session, the White Cell sends it).

**ANY CHANNEL WORKS**

Email, chat, Teams, a text file on removable media. Extra text around a pasted code is ignored, so players can add a note.

Each player's game saves after every code and appears under **RESUME**, so a game can pause for days.

# 9.0 DESIGNER'S NOTES

*Each note is keyed to the Field Manual rule it explains (DN 10.3 is about FM 10.3). It says what the rule models, how I arrived at it, and where it comes from.*

FROM RESEARCH A published source shaped this rule.

DESIGNER'S ASSUMPTION My own design choice, made so the game plays well.

THE BOARD: FIVE AREAS OF ONE MIND

| NOTICE | INTERPRET | DECIDE | ACT | TRUST |
|---|---|---|---|---|
| See events | Read them | Choose | Follow through | Be believed |
| Lose it: the other mind goes dark | Lose it: every move costs +1 | Lose it: one move fewer | Lose it: 1 in 4 moves misfire | Lose it: statements count half |

## DN 2.1 THE MIND AS TERRAIN

**MODELS** A nation's decision-making as the ground being fought over.

Two brains of about thirty connected nodes. Every node is a real kind of person or institution: a newsroom, an analysts' cell, a cabinet, a crisis team, the public's trust in its own government.

> **HOW I GOT THERE** *Cognitive warfare is hard to teach because you can't see it, and people picture it as "fake news". I wanted the real target on the table: whether the other side can still decide well under pressure.*

RUSHING, HERSCH & XU (2026)

## DN 5.4 LOSING AN AREA

**MODELS** What it costs to lose a capability.

An area falls when the other side holds at least half its nodes (DECIDE also falls if Top Leadership is held). Each loss has one plain effect (the strip above), so players feel it the way a staff would.

> **HOW I GOT THERE** *The five areas follow the research's cognitive OODA loop with trust underneath. The effects are deliberately simple so they can be read off the map.*

OODA FRAMING: RUSHING ET AL.

EFFECTS: DESIGNER'S ASSUMPTION

## DN 10.6 DECISION HEALTH

**MODELS** How well a side can still decide, 0 to 1.

The score is not views or nodes held. It is the health of the four loop areas, scaled by trust (between ×0.75 and ×1), minus a penalty for each area lost. The arithmetic is in 6.2.

> **HOW I GOT THERE** *Spreading far is not the same as changing minds. A centralized state leans 40% on DECIDE (Blue 30%): when everything runs through the top, the top is the target.*

RUSHING ET AL. · WEIGHTS AFTER SHIN (NIU)

WEIGHTS: DESIGNER'S READING

## DN 7.1 ENTRY POINTS AND FOOTHOLDS

**MODELS** Messages need a way in before they can travel.

Each brain has three entry points: News Media, Social Media and Government Contacts. Attacks start there and spread along links from nodes you have a foothold in.

> **HOW I GOT THERE** *It makes players plan an approach instead of hitting the most valuable node first, which is how I believe influence work has to be sequenced.*

DESIGNER'S ASSUMPTION

> **"SPREADING FAR IS NOT THE SAME AS CHANGING MINDS."**

# MOVES & METRICS

*Eight moves, one engagement metric, and odds that change when a side prepares.*

Plant Stories | Frame the Story | Bot Swarm | Repost & Rebrand | Watch Team | Prebunk | Official Statement | Lessons Learned

## DN 7.2 EIGHT MOVES, EIGHT CAPABILITIES

**MODELS** The attacker's and defender's capability families.

Four attack cards and four defence cards, one for each capability family in the research. Military mode keeps doctrinal names. Donovia adds a ninth, Show of Force.

> **HOW I GOT THERE** *I did not invent a list of tricks. Mapping one card to each family keeps the game honest to the framework and easy to cite in a debrief.*

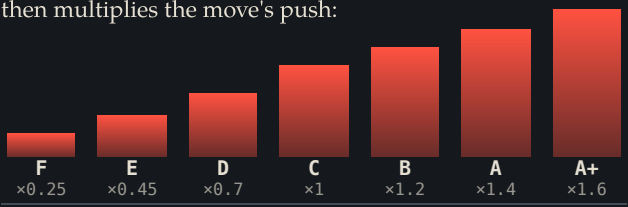
RUSHING, HERSCH & XU (2026) §2.2.2, §2.3.2

SHOW OF FORCE: ISW, IISS

## DN 7.6 THE SPREAD SCORE

**MODELS** How engaging a campaign of posts actually was.

Every attack except Show of Force rolls an audience and is graded with the research's engagement metric. The grade then multiplies the move's push:




> **HOW I GOT THERE** *I kept the formula and grade scale because they make the point that interaction outweighs views. The multipliers are mine. A viral post on the wrong area still loses the crisis.*

METRIC AND GRADES: RUSHING & XU (2025)

MULTIPLIERS: DESIGNER'S ASSUMPTION

## DN 7.5 FAILURE AND BACKFIRE

**MODELS** Platforms, watchers and prepared audiences.

Any attack can be taken down or fall flat. Watching and prebunking raise those odds, and a failed attack nudges the node back toward the defender (6.8 has the numbers).

> **HOW I GOT THERE** *I wanted defence to change the odds, not just add hit points, so players feel why detection and inoculation matter before a campaign lands.*

DESIGNER'S ASSUMPTION

## DN 9.0 RECOVERY, FRAMING AND SPREAD

**MODELS** Minds drifting back, and ideas settling in.

Each week every node recovers a little toward its own side; strongly held nodes push on their neighbours. Framing stops recovery for two weeks (three when Donovia frames), and every attack on the node lands 40% harder meanwhile: the game's version of priming.

> **HOW I GOT THERE** *Tuned so that a single hit fades but a sustained, sequenced campaign holds.*

DESIGNER'S ASSUMPTION

> **"DEFENCE SHOULD CHANGE THE ODDS, NOT JUST ADD HIT POINTS."**

# THE MOMENTS THAT COUNT

*Sixteen weeks so it fits a lunch break; four crises so preparation pays.*

| 01 | 02 ◆ TECH | 03 | 04 CRISIS | 05 | 06 ◆ TECH | 07 | 08 CRISIS | 09 | 10 ◆ TECH | 11 | 12 CRISIS | 13 | 14 ◆ TECH | 15 | 16 CRISIS ×2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

## DN 10.3 FOUR CRISES

**MODELS** Decisions under pressure at the moments that matter.

In weeks 4, 8, 12 and 16 a crisis tests one area of both minds; in the built-in campaigns the last counts double. The result depends on overall health and, heavily, on the tested area (6.4).

> **HOW I GOT THERE** *Weighting the tested area makes players read the calendar and prepare, which is the skill I wanted to reward.*

OODA FRAMING: RUSHING ET AL.

SCORING: DESIGNER'S ASSUMPTION

## DN 12.3 COGNITIVE SUPERIORITY

**MODELS** One side's decision-making clearly dominating.

If one side has lost at least two more areas than the other for two weeks running, the campaign ends early. The race shows in the side panel as LEAD.

> **HOW I GOT THERE** *The research defines cognitive superiority as a sustained advantage in the decision loop; the knockout gives that idea a hard edge.*

RUSHING, HERSCH & XU (2026)

THRESHOLDS: DESIGNER'S ASSUMPTION

## DN 11.0 NEW TECHNOLOGIES

**MODELS** Disruptive innovations arriving mid-campaign.

Both sides choose in secret: use it for an attack bonus, or prepare and deny the other side's. Using it against a prepared opponent gains nothing.

> **HOW I GOT THERE** *A dilemma, because that is the real problem with new tools: you don't know whether the other side is ready. The scoring and the threshold of 9 are the paper's.*

RUSHING, UMEOKOLO & XU (2025)

SOME SCORES: DESIGNER'S ESTIMATE

## DN 6.0 BUDGET AND MOVES

**MODELS** Finite attention and resources.

Three moves a week from a budget that grows with what you hold, capped at 18, with one extra move a turn after your last, if you can pay for it and still have 2 left. Access to officials pays more than any post.

> **HOW I GOT THERE** *The cap stops hoarding; the extra move rewards spending.*

DESIGNER'S ASSUMPTION

> **"READ THE CALENDAR, AND PREPARE THE AREA THE NEXT CRISIS TESTS."**

# RED IS NEVER A MIRROR

| | BLUE | OLVANA | DONOVIA |
|---|---|---|---|
| IS | An open democracy | A closed, one-party state | An oligarchic state at permanent information war |
| STRENGTH | Free press, truth that sticks, cheap extra move | Censored internet, state-funded content, already inside | Floods, planted premises, dormant narratives, shows of force |
| WEAKNESS | Its open internet lets Red's posts spread | Runs from the top; slow to learn | Predictable; threats can blow back |
| SHOULD FEEL | Under pressure, keeping its head | A fortress with one gate | Loud |

## DN 13.1 OPEN AGAINST CLOSED

**MODELS** Why open and closed societies are vulnerable in different ways.

Blue's posts into a censored internet spread less; Red's into Blue's open internet spread further. Blue gets trust that recovers, statements that hit harder while trust holds, and truth that sticks.

> **HOW I GOT THERE** *The asymmetry is the point: you fight a closed system differently from how it fights you.*

ATP 3-13.1 · SHIN (NIU)

SOME EFFECTS: DESIGNER'S ASSUMPTION

## DN 13.2 TWO OPPONENTS, KEPT FICTIONAL

**MODELS** Two doctrines of cognitive warfare.

Red's countries come from the Army's training world, so they are familiar and clearly fictional. Every rule names its source or is labelled as my assumption (10.1).

> **HOW I GOT THERE** *Both share a core and differ in temperament, so each feels different with a person playing it.*

ODIN DATE · NIU · EADS ET AL.

NATO C2COE · ISW · IISS · CARNEGIE

## DN 3.2 AN OPPONENT THAT PLAYS FAIR

**MODELS** A competent adversary at three levels of skill.

Recruit is noisier and earns 2 fewer points a week. Veteran plans two moves ahead on a scratch copy of the game: no extra budget, no hidden information, never the game's own dice.

> **HOW I GOT THERE** *Meant to be hard to beat, without cheating. Each of its moves also shows the likely reason, judged only from what the player could see.*

DESIGNER'S ASSUMPTION

## DN 1.3 BUILT TO BE USED

**MODELS** A teaching game that runs anywhere.

It explains itself at every moment that matters. Two players need only a turn code; campaigns are plain text that cannot run code; Immersive Briefings are optional; all text in every scheme passes 4.5:1 contrast.

> **HOW I GOT THERE** *Keep the original game: new features go around the core loop, never instead of it. The editor calls a campaign a fair match when Blue wins 25–65% of computer games; the built-in ones sit between about 35% and 60%.*

DESIGNER'S ASSUMPTION

> "MEANT TO BE HARD TO BEAT, WITHOUT CHEATING."

# HOW IT SHOULD FEEL

*Rules decide what happens; these notes are about what a player should feel while it happens. They are my intentions, not findings.*

### DN 13.3 OLVANA: A FORTRESS WITH ONE GATE

**HOW IT SHOULD FEEL** *Olvana should feel like a fortress with one gate. Against a human Olvana, Blue feels a slow squeeze: no fireworks, just patient pressure on one area at a time from someone who was already inside. Playing Olvana, you feel safe behind the firewall but know everything hangs on DECIDE, and that once it cracks you learn too slowly to fix it. It is patience against nerve.*

SHIN (NIU) · TEMPERAMENT: DESIGNER'S READING

### DN 13.4 DONOVIA: LOUD

**HOW IT SHOULD FEEL** *Donovia should feel loud. Against a human Donovia, Blue feels rushed and off balance: too many stories at once, old narratives going off like mines, a show of force just as you planned your turn. The test is whether you keep your head and defend the premises instead of chasing every claim. Playing Donovia, you feel the thrill of cheap, fast aggression, and the risk when your threats blow back.*

ISW · NATO C2COE · IISS · CARNEGIE · TEMPERAMENT: DESIGNER'S READING

### DN 15.3 IMMERSIVE BRIEFINGS: THE SECOND TELLING

**WHY IT EXISTS** *Every week in this game happens twice: once on the board, and once in how each side tells it. Immersive Briefings let you hear the second version, and hearing the other side's broadcaster tell your public that its leaders cannot decide is the point. It is optional because the base game must stay small, quiet and offline.*

DESIGNER'S CHOICE

> **"EVERY WEEK IN THIS GAME HAPPENS TWICE."**

# 10.0 SOURCES AND LIMITS

*A model is only as useful as your sense of where it is solid. This section is the honest map.*

## 10.1 WHAT COMES FROM WHERE

| RULE | FIELD MANUAL | BASIS |
|---|---|---|
| Five areas and the decision loop | 2.3 | **From research**: Rushing, Hersch & Xu (2026). |
| What losing each area does | 5.5 | **Designer's assumption.** |
| Decision Health weights | 10.6 | **Designer's reading** of Shin (2026) on centralized command. |
| The eight cards | 7.2, 8.1 | **From research**: the capability families in Rushing, Hersch & Xu (2026). Their numbers are assumptions. |
| Spread score and grades | 7.6 | **From research**: Rushing & Xu (2025). The push multipliers are assumptions. |
| Technology scores and threshold | 11.3 | **From research**: Rushing, Umeokolo & Xu (2025). 16 of the 18 score sets are the designer's estimates. |
| Crisis scoring and bands | 10.3–10.4 | **Designer's assumption.** |
| Knockout | 12.3 | Idea from Rushing, Hersch & Xu (2026); thresholds are assumptions. |
| Budget, costs, failure, recovery | 6.0, 7.5, 9.0 | **Designer's assumption.** |
| Opponent rules | 13.3, 13.4 | Each rule names its source in the game, or is marked as an assumption. |

## 10.2 WHAT HAS BEEN CHECKED

- The rules and numbers in both manuals were checked against the game's code for this version.
- Balance was tested by the computer playing itself: on Standard the built-in campaigns land between about 35% and 60% Blue wins [7.4].
- Automated tests check that every copy of a two-player or White Cell game stays in step.

## 10.3 WHAT HAS NOT

- No independent review, assurance or validation of the game, its numbers or its value for teaching.
- No study of what players learn from it.
- The cited authors and their institutions have not reviewed it.
- The computer opponent is a game opponent, not a model of any real actor's decisions.

# 10.4 REFERENCES


- Rushing, B., Hersch, W., & Xu, S. (2026). *Cognitive Warfare: Definition, Framework, and Case Study.* arXiv:2603.05222.
- Rushing, B., & Xu, S. (2025). *Quantifying the Engagement Effectiveness of Cyber Cognitive Attacks: A Behavioral Metric for Disinformation Campaigns.* arXiv:2510.15805.
- Rushing, B., Umeokolo, M.-R., & Xu, S. (2025). *Towards Proactive Defense Against Cyber Cognitive Attacks.* arXiv:2510.15801.
- Headquarters, Department of the Army (2018). *ATP 3-13.1, The Conduct of Information Operations.*
- NATO Science & Technology Organization. *Cognitive Warfare: NATO Chief Scientist Research Report.*
- Shin, D. W. (2026). *Viewing China's Cognitive Warfare Through the Lens of Its Global Strategy.* National Intelligence University.
- Eads, L. J., Clarke, R., Lin, X. S., & McCreight, R. (2023). *Warfare in the Cognitive Age: NeuroStrike and the PLA's Advanced Psychological Weapons & Tactics.*
- NATO Command and Control Centre of Excellence (2026). *Information Operations in Russian Command and Control* (short read).
- Bugayova, N., & Stepanenko, K. (2025). *A Primer on Russian Cognitive Warfare.* Institute for the Study of War.
- Voo, J., & Singh, V. V. (2025). *Russia's Information Confrontation Doctrine in Practice (2014–Present): Intent, Evolution and Implications.* International Institute for Strategic Studies.
- Wilde, G., & Sherman, J. (2023). *No Water's Edge: Russia's Information War and Regime Security*. Carnegie Endowment for International Peace.
- U.S. Army TRADOC G-2. *Decisive Action Training Environment (DATE)*, Operational Environment Data Integration Network (ODIN): Olvana, Donovia, Bothnia, Framland, Arnland, Gorgas. odin.t2com.army.mil/DATE
- Roozenbeek, J., & van der Linden, S. (2020). *Breaking Harmony Square: A game that inoculates against political misinformation* (cited in Rushing et al.).


SOURCES


### ATTRIBUTION AND LICENSE

This game and manual adapt concepts, the capability taxonomy, the engagement-effectiveness metric (formula, interaction weights and grade scale) and two sets of innovation attribute scores from the three Rushing et al. papers above. Each paper is licensed under Creative Commons Attribution 4.0 International (CC BY 4.0): creativecommons.org/licenses/by/4.0. Papers: arxiv.org/abs/2603.05222 · arxiv.org/abs/2510.15805 · arxiv.org/abs/2510.15801.

**Changes made:** the material was adapted into game mechanics and rewritten for play. Engagement grades are mapped to effect multipliers invented for this game; the OODA effects, capability families and cognitive superiority definition are expressed as game rules; 16 of the 18 technology score sets are the designer's own estimates (the paper scores AI Synthetic Media and Humanoid Robots); all scenario, narrative and tutorial text is original.

**No endorsement:** the authors and their institutions have not reviewed or endorsed this game.

**Immersive Briefings voice pack** (optional, separate download): Kokoro-82M by hexgrad, kokoro-js and transformers.js (Apache 2.0); ONNX Runtime (MIT); eSpeak NG (GPL-3.0-or-later), whose source is published with the pack.


### HOW THIS GAME WAS MADE

**Design and direction:** Kevin Williamson. Cognitive Terrain is my design: I conceived the game, built its rules from the research, chose every source, defined both opponents and the campaign briefs, set the balance targets and accessibility standards, and reviewed every rule, campaign and page.

**Tools:** I used AI-assisted development tools to help implement the code, draft text and run balance tests, working from my specifications; I reviewed, revised or rejected everything they produced. The published sources, not the tools, are the authority for any rule that models real-world behaviour. The art, icons and music are drawn and synthesized by the game's own code; no AI image or music generators were used, and the base game runs no AI while you play. The optional Immersive Briefings voice is synthetic speech from an open-source model that runs on your own computer (Field Manual 15.3).

Olvana and Donovia, and the countries Bothnia, Framland, Arnland and Gorgas, are fictional training countries from the US Army's ODIN DATE environment; they are used here without endorsement. All eight scenarios, their towns, outlets, people and every event in them are original fiction. Any resemblance to real places or events is coincidental. Donovia's rules follow the NATO, ISW, IISS and Carnegie research above; game effects are abstractions of broad, publicly reported patterns, not claims about any specific operation.

APPENDIX A · ONE PAGE · PRINT IT

# FACILITATOR'S CHECKLIST

### BEFORE

- [ ] Play the campaign yourself, end to end [3.0].
- [ ] Pick campaign, difficulty and plain or military terms [2.2].
- [ ] Put the same version on every machine [8.0].
- [ ] Test the projector scheme and text size [2.3].
- [ ] Load and test the voice pack, if used.
- [ ] Print the player aid, one per player or pair.
- [ ] White Cell: plan two or three injects [4.11].
- [ ] Decide: week in review on or off.

### DURING

- [ ] Open with the decision loop (FM 2.1) in five minutes.
- [ ] First game: tutorial on.
- [ ] Point players at the top bar's next crisis.
- [ ] When the other side pulls ahead, read its edge panel together.
- [ ] Note turning points as they happen.
- [ ] Two players: check health bars match after each code.
- [ ] White Cell: say why, in the ruling or note.
- [ ] Keep time; leave a third for the debrief.

### AFTER

- [ ] Open the debrief together [5.0].
- [ ] Compare points with the three verdicts.
- [ ] Step the replay to two or three turning points.
- [ ] Ask two questions from Appendix B.
- [ ] Ask what would differ in the real world, and why the model may be wrong.
- [ ] Keep a screenshot of each debrief.

### TIMING GUIDE

A solo campaign takes 30–45 minutes; a paired two-player game about an hour. Plan the debrief at a third of the session [2.1].

# DEBRIEF QUESTIONS

*Two starters for each campaign, tied to its twist and crises. Use them with the general questions in 5.3.*

| CAMPAIGN | QUESTIONS |
|---|---|
| **Typhoon Ines** | The Relief Flotilla tested NOTICE. What did each side need people to see first, and who got there?<br>Who was trusted with the rebuilding by week 16, and what earned it? |
| **The Valmora Vote** | Both TRUST areas started weak. What did that do to Official Statements, and how did you adapt?<br>The Leaked Audio tested INTERPRET. What would have helped people judge it? |
| **Collision at Varrin Reef** | Both sides' Government Contacts started weak. Who reached the officials first, and what did it pay?<br>Two Videos tested NOTICE. How did each side compete over which clip people saw? |
| **Fever in Port Sabre** | Rumours spread faster here. When did a cluster of held nodes start doing the work for one side?<br>The Bad Batch tested TRUST. What protects trust in health advice before a scandal, not after? |
| **Winter Grid** | Four dormant narratives were hidden in Blue's mind. When did you find the first, and how?<br>The Price of Heat tested DECIDE. What made the decision hard to take? |
| **The Lake Referendum** | Donovia began with a foothold in Blue's INTERPRET. What premise had it planted, and did Blue answer the premise or the stories?<br>Debate Night tested DECIDE. Who set the terms of the debate? |
| **The Silent Survey Ship** | Both DECIDE areas started strained. Who acted first under uncertainty, and was acting first right?<br>Three Stories tested INTERPRET. How did competing explanations change the decision? |
| **The Vell Valley** | The Protection Claim tested INTERPRET. How did Donovia's framing shape what the valley's events meant?<br>Did a Show of Force work, or blow back? What decided it? |

**FOR ANY CAMPAIGN**

- Which rule in this game would you change to make it more like the real world? What would that change in how it plays?
- Where did the game's numbers feel wrong to you? (Section 10.0 says which are assumptions.)

APPENDIX C · SHARED WITH THE FIELD MANUAL

# REFERENCE TABLES

*The player-facing numbers in one place, as in Field Manual Appendix A. Rule numbers refer to the Field Manual.*

## CARDS [FM 7.2] [FM 8.1] [FM 13.4]

| CARD | COST | PUSH | COVERS |
|---|---|---|---|
| PLANT STORIES | 2 | 20 (×1.5 on entry) | Target |
| FRAME THE STORY | 2 | 16; framed | Target |
| BOT SWARM | 3* | 12; 7 on links | Target + links |
| REPOST & REBRAND | 2 | 10; strips watched, prebunked | Target; strips on links too |
| SHOW OF FORCE | 3 | 16 (no grade) | Any ACT/DECIDE |
| WATCH TEAM | 2 | Watched 3 wk | Target + links |
| PREBUNK | 2 | Prebunked 3 wk | Target + links |
| OFFICIAL STATEMENT | 2 | 14 back (Blue 22) | Target |
| LESSONS LEARNED | 3 | +4 to all (Red +2) | Whole mind |

* 2 for Red. Add 1 to every cost while INTERPRET is lost.

## FAILURE [FM 7.5]

| | |
|---|---|
| Attack | 8%, +15% watched, +8% prebunked |
| Blue into Olvana / Donovia | +2% / +1% |
| Show of Force | +4% |
| Official Statement | 6%; 26% if TRUST lost |
| Any move, ACT lost | 1 in 4 misfires |

## GRADES [FM 7.6]

| SCORE | GRADE | PUSH × |
|---|---|---|
| under 3 | F | 0.25 |
| 3 to under 4 | E | 0.45 |
| 4 to under 10 | D | 0.7 |
| 10+ | C | 1 |
| 100+ | B | 1.2 |
| 1,000+ | A | 1.4 |
| 10,000+ | A+ | 1.6 |

## CRISES [FM 10.3] [FM 10.4]

Score = 0.6 × Decision Health + 0.4 × tested area, −0.08 if that area is lost

| | |
|---|---|
| SOUND | 0.86 or above: 3 points |
| DELAYED | 0.72 to under 0.86: 1 point |
| ERROR | under 0.72: 0, other side +1 |

## WEEKLY RECOVERY [FM 9.1]–[FM 9.3]

| | |
|---|---|
| Recovery | 1 + Lessons level (Red +0.5 each) |
| Support | +1 if a link leans 50+ your way |
| Spread | 5 per link leaning 50+ the other way |

## THRESHOLDS [FM 5.2]–[FM 5.4] [FM 6.5]

| | | | |
|---|---|---|---|
| Foothold | 10 | Held | 30 |
| Area lost | half its nodes | Budget cap | 18 |